BirdNET ONNX Export

Status update · What we proved, what was found, what's next

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The Model Port Is Done and Proven

We re-exported BirdNET from TensorFlow into ONNX — a portable format that runs anywhere without TF — and then proved it produces identical predictions to the original.

The conversion tool hit a wall — and I had to rewrite part of BirdNET's internal audio math into an algebraically identical form to get past it. Verified before converting anything.

60,243

clips validated
across all 6 recorders

100%

agreement with the
original model

A Data Mystery — and How We Cracked It

Agreement dropped to 0% on some recorders

Expanding from one recorder to all six, some came back with zero agreement. Looked like a bug.

Three-way check ruled out our side entirely

Preprocessing was bit-exact. Original TF model agreed with our ONNX. Both disagreed with the reference — so the reference was the mismatch.

Root cause: wrong clip lengths in the shared audio

The reference CSV expects exactly 3.000 s clips. Most shared audio was 2.987 s or 3.072 s — a different segmentation run. Qian re-cut it; all six recorders now score 100%.

Compression: Half the Size, Nearly Free

We also produced a compressed (FP16) version — half the file size.

FP16 came back at 99.85% agreement — suspiciously high. So we dug in rather than celebrating. Turns out the runtime computes in FP32 anyway on CPU, so we're measuring storage precision only, not full FP16 arithmetic.

Qian confirmed: her 95.3% (TFLite) and our 99.85% (ONNX) are both valid — just different tools rounding weights differently. Neither contradicts the other.

31.2 MB

vs 59.4 MB FP32
~half the size

99.854%

agreement on CPU runtime

every disagreement = two owls being confused for each other

What's Next

WAITING ON

Hardware decision (Kyle)

The deployment hardware isn't locked yet. Once it is, we measure real-world speed and memory — and it also decides how we read the FP16 number.

IN PROGRESS

Class pruning — fix before build

Reducing BirdNET from 6,522 species to the regional set. Need to fix the species-list basis first — the pruning plan was built against the wrong label file.

SOON

Pack contract with Daan

Defining the schema for how BirdNET ships inside the DDP pack. The export work directly feeds this — we now know the real I/O contract, the patch step, and what can change.